

Supplementary Information: Posterior Information Distortion, Effective Sample Size, Corrected Covariance, and Bias

1. Log-Posterior and Score Definitions

Consider a trajectory of T sequential measurement intervals with data $D = (y_1, \dots, y_T)$, and a Gaussian prior $\pi(\theta)$ on the parameter vector $\theta \in \mathbb{R}^d$. Throughout this supplement, all derivatives, scores, and Hessians are taken in the transformed coordinates in which π is Gaussian; no Jacobian corrections appear. We restrict to Gaussian priors with fixed (non-random) hyperparameters, so the prior log-density and its derivatives are deterministic functions of θ only. As in `supplement_information_distortion_main.tex`, all expectations and covariances are taken under the frequentist sampling distribution of D at a fixed true θ (the role of the prior here is to regularize curvature and shift the score mean, not to randomize θ).

Notation bridge to the Likelihood supplement. We adopt the subscripts `lik`, `prior`, `post` to distinguish the data, prior, and combined contributions. The likelihood-side objects are renamings of the unsubscripted objects of the Likelihood supplement:

$$\mathbf{H}_{\text{lik}} \equiv \mathbf{H}, \quad \mathbf{J}_{\text{T,lik}} \equiv \mathbf{J}_{\text{total}}, \quad \mathbf{J}_{\text{S,lik}} \equiv \mathbf{J}_{\text{sample}}, \quad S_{\text{lik}} \equiv S, \quad (1)$$

all of Sections 1–3 of `supplement_information_distortion_main.tex`. We also write $\ell(\theta) = \log L(\theta)$ as in the Likelihood supplement.

Log-posterior and posterior score. The log-posterior decomposes as

$$\log p(\theta \mid D) = \ell(\theta) + \log \pi(\theta) + \text{const}, \quad (2)$$

where $\ell(\theta) = \sum_{t=1}^T \ell_t(\theta)$ with $\ell_t(\theta) = \log p(y_t \mid \theta, y_{1:t-1})$, and the additive constant $-\log p(D)$ does not depend on θ . The posterior score is

$$S_{\text{post}}(\theta) = \nabla_{\theta} \log p(\theta \mid D) = S_{\text{lik}}(\theta) + s_{\pi}(\theta), \quad (3)$$

where

$$S_{\text{lik}}(\theta) = \sum_{t=1}^T s_t(\theta), \quad s_t(\theta) := \nabla_{\theta} \ell_t(\theta), \quad s_{\pi}(\theta) := \nabla_{\theta} \log \pi(\theta). \quad (4)$$

For the Gaussian prior $\pi(\theta) = \mathcal{N}(\mu_{\text{prior}}, \Sigma_{\text{prior}})$ with precision $\mathbf{H}_{\text{prior}} = \Sigma_{\text{prior}}^{-1}$, this gives the explicit form

$$s_{\pi}(\theta) = -\mathbf{H}_{\text{prior}}(\theta - \mu_{\text{prior}}). \quad (5)$$

Per-interval vs. global decomposition. The likelihood score retains the per-interval structure inherited from the trajectory model, while the prior contributes a single global term that is not indexed by measurement interval. We write

$$S_{\text{post}}(\theta) = \left(\sum_{t=1}^T s_t(\theta) \right) + s_{\pi}(\theta), \quad (6)$$

distinguishing the per-interval sum from the global prior term.

Prior as a global deterministic observation. Throughout this supplement we treat the prior $\pi(\theta)$ as a single global “observation” contributing one deterministic score $s_{\pi}(\theta)$ and one deterministic curvature $\mathbf{H}_{\text{prior}} = -\nabla_{\theta}^2 \log \pi(\theta)$. This convention has the following consequences:

- At fixed θ , $s_{\pi}(\theta)$ is a constant under the data-sampling distribution; hence its sampling variance vanishes and its sampling covariance with any $s_t(\theta)$ is zero.
- The prior contributes additively to the curvature budget without introducing cross-interval terms.
- For a Gaussian prior, $\mathbf{H}_{\text{prior}}$ is constant in θ and diagonal in the transformed coordinates.
- When $\mathbf{H}_{\text{prior}}$ is positive definite (proper Gaussian prior) and $\mathbf{H}_{\text{lik}} \succeq 0$, the posterior curvature $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ developed in Section 2 is strictly positive definite, so the inverse square roots used in Sections 3–4 are well-defined without subspace projection (cf. Section 8 of the Likelihood supplement).

Posterior score mean and covariance under sampling. Combining (3) with the deterministic-prior-score convention,

$$\mathbb{E}[S_{\text{post}}(\theta)] = \mathbb{E}[S_{\text{lik}}(\theta)] + s_{\pi}(\theta), \quad (7)$$

$$\text{Cov}(S_{\text{post}}(\theta)) = \text{Cov}(S_{\text{lik}}(\theta)) = \mathbf{J}_{\text{T,lik}}, \quad (8)$$

i.e. the prior shifts the mean of the posterior score but contributes nothing to its sampling covariance. The covariance contribution of the prior enters the diagnostic framework only through its curvature $\mathbf{H}_{\text{prior}}$, as defined in Section 3.

Important: $\mathbf{J}_{\text{T,post}}$ is NOT $\text{Cov}(S_{\text{post}})$. Section 3 will define $\mathbf{J}_{\text{T,post}} := \mathbf{J}_{\text{T,lik}} + \mathbf{H}_{\text{prior}}$ by treating the prior as a virtual observation whose deterministic score contributes information $\mathbf{H}_{\text{prior}}$ to the curvature/fluctuation budget. By (8), this is *not* the literal covariance of S_{post} : $\mathbf{J}_{\text{T,post}} \neq \text{Cov}(S_{\text{post}})$. This definitional choice is what makes the curvature/fluctuation identity $\mathbf{J}_{\text{T,post}} = \mathbf{H}_{\text{post}}$ hold under correct likelihood specification regardless of prior strength.

Reduction to the likelihood case. When the prior becomes diffuse $\mathbf{H}_{\text{prior}} \rightarrow 0$ (formally an improper uniform prior, with $s_{\pi} \rightarrow 0$ as an automatic consequence of (5) at any fixed θ), equations (3)–(8) collapse to the corresponding likelihood definitions and $S_{\text{post}} \rightarrow S_{\text{lik}}$. When $\mathbf{H}_{\text{lik}}$ is full rank, the limit is regular and every Posterior IDM quantity defined in subsequent sections reduces smoothly to its Likelihood counterpart. When $\mathbf{H}_{\text{lik}}$ is singular, the limit recovers the subspace-projection regime of Section 8 of the Likelihood supplement; the regularizing role of a strictly positive $\mathbf{H}_{\text{prior}}$ is exactly what Posterior IDM exploits to avoid that projection.

2. Posterior Hessian Decomposition

The negative log-posterior decomposes as

$$-\log p(\theta \mid D) = -\ell(\theta) - \log \pi(\theta) + \text{const}, \quad (9)$$

so its expected curvature splits additively,

$$\mathbf{H}_{\text{post}}(\theta) = \mathbf{H}_{\text{lik}}(\theta) + \mathbf{H}_{\text{prior}}(\theta). \quad (10)$$

The likelihood term $\mathbf{H}_{\text{lik}}$ is the Gaussian Fisher Information of Section 2 of the Likelihood supplement:

$$\mathbf{H}_{\text{lik}}(\theta) = \sum_{t=1}^T \mathbb{E}[\mathbf{I}_t^{\text{G}}(\theta)] = -\mathbb{E}[\nabla_{\theta}^2 \ell(\theta)], \quad (11)$$

where the second equality is the Fisher identity, valid under correct likelihood specification and standard regularity (interchange of differentiation and expectation). The diagnostic role of $\mathbf{H}_{\text{lik}}$ versus $\mathbf{J}_{\text{T,lik}}$ (Section 3 below) is to quantify departures from this identity.

The prior term is the precision matrix of the Gaussian prior in the transformed coordinates,

$$\mathbf{H}_{\text{prior}}(\theta) = -\nabla_{\theta}^2 \log \pi(\theta) = \Sigma_{\text{prior}}^{-1}, \quad (12)$$

which is constant in θ . When the prior factorizes across the transformed coordinates, $\mathbf{H}_{\text{prior}}$ is diagonal with entries $1/\sigma_{\text{prior},i}^2$.

The matrix $\mathbf{H}_{\text{post}}$ defines the local quadratic geometry of the posterior at the MAP and reduces to $\mathbf{H}_{\text{lik}}$ in the flat-prior limit $\mathbf{H}_{\text{prior}} \rightarrow 0$ (Section 2.3).

2.1 Positive-Definiteness of $\mathbf{H}_{\text{post}}$

A proper Gaussian prior has $\mathbf{H}_{\text{prior}} \succ 0$. When the analytic Gaussian Fisher Information (11) is used, $\mathbf{H}_{\text{lik}} \succeq 0$ by construction (as a sum of expected outer-product terms). Then for any nonzero $v \in \mathbb{R}^d$,

$$v^{\top} \mathbf{H}_{\text{post}} v \geq v^{\top} \mathbf{H}_{\text{prior}} v \geq \lambda_{\min}(\mathbf{H}_{\text{prior}}) \|v\|^2 > 0, \quad \mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{prior}} \succ 0, \quad (13)$$

so $\mathbf{H}_{\text{post}}$ is strictly positive definite even when $\mathbf{H}_{\text{lik}}$ is singular or rank-deficient. No subspace projection is required.

Indefinite estimator-based $\mathbf{H}_{\text{lik}}$. In practice $\mathbf{H}_{\text{lik}}$ is often approximated by a finite-difference Hessian whose estimator bias can produce negative eigenvalues along weakly identified directions. In that case the Loewner bound $\mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{prior}}$ no longer follows; instead, positive definiteness of $\mathbf{H}_{\text{post}}$ requires the prior to dominate the negative spectrum,

$$\lambda_{\min}(\mathbf{H}_{\text{prior}}) > -\lambda_{\min}(\mathbf{H}_{\text{lik}}), \quad (14)$$

which is a mild and easily monitored condition for proper priors and bounded estimator error.

2.2 Consequences for Posterior IDM

Equation (13) has two structural consequences that distinguish Posterior IDM from its Likelihood counterpart:

- All Posterior IDM quantities defined below ($\mathbf{C}_{\text{post}}$, $\mathbf{C}_{\text{sample,post}}$, $\mathbf{R}_{\text{post}}$, $\Sigma_{\text{DCC,post}}$, $\mathbf{b}_{\text{DIB,post}}$) are well-defined on the full parameter space whenever $\mathbf{H}_{\text{lik}} \succeq 0$ (analytic Gaussian Fisher case) or whenever (14) holds (finite-difference case). The subspace projection of Section 8 of the Likelihood supplement is unnecessary in either case.
- The factorization required by the whitening transforms is obtained by a single Cholesky factorization $\mathbf{H}_{\text{post}} = \mathbf{L}_{\text{post}}\mathbf{L}_{\text{post}}^\top$ followed by triangular solves. The result is similar (by a congruence) to the symmetric whitening $\mathbf{H}_{\text{post}}^{-1/2} \cdot \mathbf{H}_{\text{post}}^{-1/2}$ and preserves all spectral invariants (eigenvalues, trace, determinant) but does not coincide with the symmetric form as a matrix; see Section 4.5 for the implications for per-parameter diagnostics.

In computations we never form $\mathbf{H}_{\text{lik}}$ and $\mathbf{H}_{\text{prior}}$ separately when a stable factor of $\mathbf{H}_{\text{post}}$ is required: we assemble the sum (10) first, then factor.

2.3 Limiting Regimes

Flat-prior limit. As $\mathbf{H}_{\text{prior}} \rightarrow 0$, (10) degenerates to $\mathbf{H}_{\text{post}} \rightarrow \mathbf{H}_{\text{lik}}$. Combined with $\mathbf{J}_{\text{T,post}} \rightarrow \mathbf{J}_{\text{T,lik}}$ and $\mathbf{J}_{\text{S,post}} \rightarrow \mathbf{J}_{\text{S,lik}}$ from Section 3, every Posterior IDM quantity reduces termwise to its Likelihood IDM counterpart. When $\mathbf{H}_{\text{lik}}$ is rank-deficient, this collapse is understood on the retained eigenspace of $\mathbf{H}_{\text{lik}}$ (the singular-Fisher machinery of Section 8 of the Likelihood supplement re-emerges as the appropriate generalization).

Informative-prior limit. In the opposite limit $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \rightarrow \infty$, $\mathbf{H}_{\text{post}}$ is dominated by $\mathbf{H}_{\text{prior}}$, the posterior contracts to the prior, and the conditioning of $\mathbf{H}_{\text{post}}$ is set by the prior precision. All Posterior IDM quantities remain well-defined throughout this entire range.

3. Posterior Score Fluctuation Matrices

The diagnostic role played by $\mathbf{J}_{\text{T,lik}}$ ($\equiv \mathbf{J}_{\text{total}}$) and $\mathbf{J}_{\text{S,lik}}$ ($\equiv \mathbf{J}_{\text{sample}}$) for the likelihood (Section 3 of the Likelihood supplement) must be lifted to the posterior so that the comparison with $\mathbf{H}_{\text{post}}$ yields meaningful values.

Recall the posterior score (3). Taken over realizations of D at fixed θ , $s_\pi(\theta)$ is a constant; hence it has zero sampling variance and, as a consequence, zero sampling covariance with $S_{\text{lik}}(\theta)$. The raw posterior score covariance therefore satisfies

$$\text{Cov}(S_{\text{post}}(\theta)) = \text{Cov}(S_{\text{lik}}(\theta)) = \mathbf{J}_{\text{T,lik}}, \quad (15)$$

i.e. it does *not* contain $\mathbf{H}_{\text{prior}}$, while the negative posterior Hessian $\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}$ does. A direct comparison $\mathbf{H}_{\text{post}}^{-1/2} \mathbf{J}_{\text{T,lik}} \mathbf{H}_{\text{post}}^{-1/2}$ would conflate prior shrinkage with genuine information distortion and would not equal $\mathbf{I}$ under correct likelihood specification.

3.1 Prior as a single deterministic observation

We adopt the convention of treating the prior as one additional observation, independent of the data, with precision $\mathbf{H}_{\text{prior}}$ and a score that is deterministic across data realizations (zero fluctuation in D at fixed θ). Under this convention, the prior contributes its precision to every information-style matrix but contributes no temporal cross-terms with the data, since it is fixed before D is observed. We accordingly define

$$\mathbf{J}_{\text{T,post}} := \mathbf{J}_{\text{T,lik}} + \mathbf{H}_{\text{prior}}, \quad (16)$$

$$\mathbf{J}_{\text{S,post}} := \mathbf{J}_{\text{S,lik}} + \mathbf{H}_{\text{prior}}. \quad (17)$$

$\mathbf{J}_{\text{T,post}}$ is a defined object, not a covariance. By (15), $\mathbf{J}_{\text{T,post}}$ is *not* equal to the literal covariance of $S_{\text{post}}(\theta)$ (which is $\mathbf{J}_{\text{T,lik}}$). Definition (16) is the augmented-observation bookkeeping convention that places the prior on the same footing as the data: it contributes its curvature directly to the information budget rather than through sampling variance. This convention is what propagates the likelihood Bartlett identity to the posterior in (18) below.

Inheritance of the information identity. With (16), the posterior information identity inherits from the likelihood one. If $\mathbf{J}_{\text{T,lik}} = \mathbf{H}_{\text{lik}}$ (correct likelihood specification, Section 4 of the Likelihood supplement), then

$$\mathbf{J}_{\text{T,post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}} = \mathbf{H}_{\text{post}}. \quad (18)$$

This identity holds by construction of $\mathbf{J}_{\text{T,post}}$ (Eq. (16)), not as an independent Bartlett identity for the posterior score; it is the additive convention that propagates the likelihood identity. Crucially, the equality survives arbitrary rescaling of $\mathbf{H}_{\text{prior}}$: as the prior ranges from weakly to dominantly informative, $\mathbf{J}_{\text{T,post}}$ and $\mathbf{H}_{\text{post}}$ change in lockstep, so the resulting $\mathbf{C}_{\text{post}} = \mathbf{I}$ in Section 4 holds regardless of prior strength.

3.2 Sample/total decomposition

There is no posterior per-interval score $s_{t,\text{post}}$; the prior is treated as a single extra observation, so the within-interval covariance sum in $\mathbf{J}_{\text{S,post}}$ is over data intervals only. Since the prior pseudo-score is deterministic, $\text{Cov}(s_t, s_\pi) = 0$ for every data interval t and the cross-interval sum runs only over data indices. The decomposition of $\mathbf{J}_{\text{T,lik}}$ into within-interval variance plus cross-interval covariance therefore lifts unchanged:

$$\mathbf{J}_{\text{T,post}} = \mathbf{J}_{\text{S,post}} + 2 \sum_{i < j} \text{Cov}(s_i, s_j), \quad i, j \in \{1, \dots, T\}, \quad (19)$$

with s_t the per-interval data score of (4). The prior contributes $\mathbf{H}_{\text{prior}}$ identically to both $\mathbf{J}_{\text{T,post}}$ and $\mathbf{J}_{\text{S,post}}$, and therefore drops out of the difference $\mathbf{J}_{\text{T,post}} - \mathbf{J}_{\text{S,post}}$, which equals $2 \sum_{i < j} \text{Cov}(s_i, s_j)$ and involves only the data scores. Equation (19) isolates temporal dependence in the data score from the prior contribution, exactly as in the likelihood case.

3.3 Vanishing prior limit

As $\mathbf{H}_{\text{prior}} \rightarrow \mathbf{0}$,

$$\mathbf{J}_{\text{T,post}} \rightarrow \mathbf{J}_{\text{T,lik}}, \quad \mathbf{J}_{\text{S,post}} \rightarrow \mathbf{J}_{\text{S,lik}}, \quad \mathbf{H}_{\text{post}} \rightarrow \mathbf{H}_{\text{lik}}, \quad (20)$$

and all posterior distortion matrices defined in Section 4 reduce to their Likelihood IDM counterparts. When $\mathbf{H}_{\text{lik}}$ is full rank the limit is direct; when $\mathbf{H}_{\text{lik}}$ is rank-deficient, the limit must be interpreted on its retained eigenspace (Section 8 of the Likelihood supplement). For strictly positive $\mathbf{H}_{\text{prior}}$, by contrast, the Posterior IDM is well-defined globally without any retained-subspace construction; this is the structural advantage exploited in Section 4.

4. Posterior Information Distortion (PID)

The central Bayesian diagnostic is the total Posterior Information Distortion

$$\mathbf{C}_{\text{post}}(\theta) = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{J}_{\text{T,post}} \mathbf{H}_{\text{post}}^{-1/2}, \quad (21)$$

the direct counterpart of the Likelihood Information Distortion $\mathbf{C} = \mathbf{H}^{-1/2} \mathbf{J}_{\text{total}} \mathbf{H}^{-1/2}$ (Section 4 of the Likelihood supplement; recall the bridge (1) that identifies $\mathbf{H} \equiv \mathbf{H}_{\text{lik}}$ and $\mathbf{J}_{\text{total}} \equiv \mathbf{J}_{\text{T,lik}}$), with the likelihood curvature and total score covariance replaced by their posterior counterparts

$$\mathbf{H}_{\text{post}} = \mathbf{H}_{\text{lik}} + \mathbf{H}_{\text{prior}}, \quad \mathbf{J}_{\text{T,post}} = \mathbf{J}_{\text{T,lik}} + \mathbf{H}_{\text{prior}}. \quad (22)$$

Throughout, $\mathbf{H}_{\text{lik}}$, $\mathbf{H}_{\text{prior}}$, $\mathbf{J}_{\text{T,lik}}$, and $\mathbf{J}_{\text{T,post}}$ are symmetric, and the symmetric inverse square root $\mathbf{H}_{\text{post}}^{-1/2}$ is the unique positive definite root of $\mathbf{H}_{\text{post}}^{-1}$.

4.1 Consistency under correct likelihood specification

Under correct likelihood specification and standard regularity at the true parameter θ_0 , the population information identity $\mathbf{J}_{\text{T,lik}} = \mathbf{H}_{\text{lik}}$ holds; empirical plug-ins satisfy it only up to sampling fluctuation. Substituting into (22) gives (18), $\mathbf{J}_{\text{T,post}} = \mathbf{H}_{\text{post}}$, and therefore

$$\mathbf{C}_{\text{post}} = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{H}_{\text{post}} \mathbf{H}_{\text{post}}^{-1/2} = \mathbf{I}. \quad (23)$$

This holds for *any* proper Gaussian prior, informative or weak: the prior contribution $\mathbf{H}_{\text{prior}}$ enters $\mathbf{H}_{\text{post}}$ and $\mathbf{J}_{\text{T,post}}$ symmetrically and cancels under the whitening transform.

The companion diagnostics $\mathbf{C}_{\text{sample,post}} = \mathbf{H}_{\text{post}}^{-1/2} \mathbf{J}_{\text{S,post}} \mathbf{H}_{\text{post}}^{-1/2}$ and $\mathbf{R}_{\text{post}} = \mathbf{J}_{\text{S,post}}^{-1/2} \mathbf{J}_{\text{T,post}} \mathbf{J}_{\text{S,post}}^{-1/2}$ play the same role for within-interval mismatch and temporal correlation as $\mathbf{C}_{\text{sample}}$ and $\mathbf{R}$ do in the Likelihood supplement. Under correct specification with no temporal correlation, $\mathbf{C}_{\text{post}} = \mathbf{C}_{\text{sample,post}} = \mathbf{R}_{\text{post}} = \mathbf{I}$; under correct specification with temporal correlation, $\mathbf{C}_{\text{sample,post}} = \mathbf{I}$ and $\mathbf{R}_{\text{post}} \neq \mathbf{I}$, with $\mathbf{C}_{\text{post}}$ tracking $\mathbf{R}_{\text{post}}$ (the formal development is deferred to a separate section in the spirit of Section 10 of the Likelihood supplement).

4.2 Deviation as a pure likelihood diagnostic

If the likelihood is misspecified, $\mathbf{J}_{\text{T,lik}} \neq \mathbf{H}_{\text{lik}}$ and

$$\mathbf{J}_{\text{T,post}} - \mathbf{H}_{\text{post}} = \mathbf{J}_{\text{T,lik}} - \mathbf{H}_{\text{lik}}, \quad (24)$$

so the prior contribution drops out of the unwhitened residual. Equivalently,

$$\mathbf{C}_{\text{post}} - \mathbf{I} = \mathbf{H}_{\text{post}}^{-1/2} (\mathbf{J}_{\text{T,lik}} - \mathbf{H}_{\text{lik}}) \mathbf{H}_{\text{post}}^{-1/2}. \quad (25)$$

The numerator $\mathbf{J}_{\text{T,lik}} - \mathbf{H}_{\text{lik}}$ depends only on likelihood quantities, so any nonzero deviation $\mathbf{C}_{\text{post}} \neq \mathbf{I}$ originates purely on the likelihood side: the prior cannot create distortion of this kind. The

whitening matrix $\mathbf{H}_{\text{post}}^{-1/2}$ does, however, rescale the magnitude of the deviation: a strong prior shrinks $\mathbf{C}_{\text{post}} - \mathbf{I}$ toward zero (in the limit $\lambda_{\min}(\mathbf{H}_{\text{prior}}) \rightarrow \infty$, $\mathbf{C}_{\text{post}} \rightarrow \mathbf{I}$), while a weak prior recovers the Likelihood IDM magnitude $\mathbf{C} - \mathbf{I} = \mathbf{H}_{\text{lik}}^{-1/2}(\mathbf{J}_{\text{T,lik}} - \mathbf{H}_{\text{lik}})\mathbf{H}_{\text{lik}}^{-1/2}$ (Section 4.3). Thus $\mathbf{C}_{\text{post}}$ diagnoses purely likelihood-side miscalibration, with a magnitude calibrated by total posterior curvature.

4.3 Flat-prior limit

As $\mathbf{H}_{\text{prior}} \rightarrow \mathbf{0}$,

$$\mathbf{H}_{\text{post}} \rightarrow \mathbf{H}_{\text{lik}}, \quad \mathbf{J}_{\text{T,post}} \rightarrow \mathbf{J}_{\text{T,lik}}, \quad \mathbf{C}_{\text{post}} \rightarrow \mathbf{C}, \quad (26)$$

recovering the Likelihood IDM of Section 4 of the Likelihood supplement. When $\mathbf{H}_{\text{lik}}$ is full rank this limit is literal; when $\mathbf{H}_{\text{lik}}$ is rank-deficient it is understood on its retained eigenspace (Section 8 of the Likelihood supplement), and the Posterior IDM with a vanishing-but-proper prior provides a regularized version of that subspace construction. The Posterior IDM is therefore a strict generalization that reduces to its frequentist analogue in the noninformative limit.

4.4 Well-posedness without subspace projection

For any proper Gaussian prior, $\mathbf{H}_{\text{prior}} \succ \mathbf{0}$ (diagonal in transformed coordinates). Combined with $\mathbf{H}_{\text{lik}} \succeq \mathbf{0}$ for the analytic Gaussian Fisher Information, Eq. (13) gives

$$\mathbf{H}_{\text{post}} \succeq \mathbf{H}_{\text{prior}} \succ \mathbf{0}, \quad \text{so} \quad \mathbf{H}_{\text{post}}^{-1/2} \text{ in (21) is unconditionally defined.} \quad (27)$$

Hence $\mathbf{C}_{\text{post}}$ requires no retained-eigenspace construction. When $\mathbf{H}_{\text{lik}}$ is a finite-difference estimator that can develop negative eigenvalues, well-posedness instead requires the explicit dominance condition (14); for proper priors and bounded estimator error this is easily monitored. This contrasts with the Likelihood IDM, where singular $\mathbf{H}_{\text{lik}}$ forces the subspace treatment of Section 8 of the Likelihood supplement.

Numerical caveat. Mathematical well-posedness does not imply numerical robustness: directions along which both $\mathbf{H}_{\text{lik}} \approx \mathbf{0}$ and the prior precision is small produce $\mathbf{H}_{\text{post}}$ with large condition number, and $\mathbf{C}_{\text{post}}$ entries in those directions are sensitive to prior specification. We recommend reporting the condition number of $\mathbf{H}_{\text{post}}$ alongside any $\mathbf{C}_{\text{post}}$ diagnostic.

4.5 Computation

We never form $\mathbf{H}_{\text{post}}^{-1}$ explicitly. Compute the Cholesky factor

$$\mathbf{H}_{\text{post}} = \mathbf{L}_{\text{post}}\mathbf{L}_{\text{post}}^\top, \quad (28)$$

where $\mathbf{L}_{\text{post}}$ is distinct from the $\mathbf{L}$ of the Likelihood supplement (which factors $\mathbf{H}_{\text{lik}}$). The congruent matrix

$$\tilde{\mathbf{C}}_{\text{post}} = \mathbf{L}_{\text{post}}^{-1}\mathbf{J}_{\text{T,post}}\mathbf{L}_{\text{post}}^{-\top} \quad (29)$$

is obtained by two triangular solves applied to the columns of $\mathbf{J}_{\text{T,post}}$. Strictly, $\tilde{\mathbf{C}}_{\text{post}}$ is similar (by a non-orthogonal congruence) to $\mathbf{C}_{\text{post}}$ of (21), not equal to it as a matrix; the two share all *spectral* invariants (eigenvalues, trace, determinant), which is what the per-eigen-direction diagnostics of Section 5 of the Likelihood supplement use.

Per-parameter vs. spectral diagnostics. Diagonal entries are *not* preserved under congruence: $[\tilde{\mathbf{C}}_{\text{post}}]_{ii} \neq [\mathbf{C}_{\text{post}}]_{ii}$ in general. Per-parameter diagnostics $[\mathbf{C}_{\text{post}}]_{ii}$ therefore require the symmetric form (21), computed via the spectral decomposition of $\mathbf{H}_{\text{post}}$; only spectral diagnostics (eigenvalues, $\text{tr}(\mathbf{C}_{\text{post}})$, $\det(\mathbf{C}_{\text{post}})$) may be read off from the Cholesky form (29). With this distinction made, the interpretation of $\mathbf{C}_{\text{post}}$ follows the Likelihood IDM verbatim:

- $[\mathbf{C}_{\text{post}}]_{ii} = 1$ indicates posterior consistency along parameter i ; values > 1 (< 1) indicate under- (over-) estimated posterior uncertainty relative to the Bayesian curvature $\mathbf{H}_{\text{post}}$.
- $\lambda_i(\mathbf{C}_{\text{post}}) = 1$ indicates consistency along the i -th eigen-direction; deviations have the same inflation/deflation interpretation as in Section 5 of the Likelihood supplement.